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Research Assignment 1

1.] Database is an organized, electronic collection of data that is structured to allow for fast storage, retrieval, management and updates

- Handling Massive data Volume
- Data Integrity and Security

Examples

- Online Shopping / E-commerce
- Banking and financial Services

2.] Data warehouse is a centralized repository that aggregates structured data from multiple, disparate sources

Differences

Feature	Regular database (OLTP)	Data Warehouse (OLAP)
Purpose	Day to day operations / transactions	Decision Support, analysis

why Companies use data warehouses

- Improved Decision Making
- Single Source of truth

3.] A data lake is a centralized repository that store vast amounts of raw, unstructured, semi-structured and structured data in its native format until needed

Differences

- Data lake store raw, "as-is" (schema on read) whereas data warehouses store processed, structured data (schema-on-write)

Pros

- Storing diverse data types
- Advanced Analytics
- Future Proofing data

4.] Data Mart is a specialized, subject-oriented subset of a data warehouse designed for a specific team, such as marketing, finance, or sales

How it relate to data warehouses

- Dependent data marts
- Independent Data Marts
- Virtual Data Marts

Typically uses a data Mart

- Marketing department
- finance team

5.] Structured data fits neatly into a relational database management system

Example: Customer Relationship Management (CRM)

Semi structured data does not reside in a relational database but uses tags

Example: JSON (JavaScript Object Notation)

6.] Data Modeling is the process of creating a visual representation or blueprint of an entire information system, mapping out how data is organized, stored and accessed

Importance

- Data integrity and quality
- Improved performance

Types

- Conceptual data Modeling
- Logical data Modeling

7.] Data mining is the process of analyzing large, complex datasets to identify hidden patterns, trends, correlations and anomalies

Used in business

Businesses use data mining to turn vast amounts of

accumulated data

Examples

- E-commerce product recommendations (Amazon)
 - Financial fraud detection (Banking Sector)
- 8] Database Management System is software that acts as an interface between users, applications and the data stored in a database

Oracle database

A feature rich, high performance and secure enterprise grade relational DBMS

MySQL

An open source relational database management system often used for web and application development

9] ELT (Extract, Load, Transform) is a modern data integration process that reverses the traditional ETL order by loading raw data directly into a target repository (such as a cloud data warehouse) before transforming it

Steps

1. Extract (E)

what it does → data is gathered from various sources such as SaaS applications

2. Load (L)

what it does → The raw, extracted data is immediately moved into a central data repository, such as a cloud data warehouse

3. Transform (T)

what it does → the data is cleansed, validated, organized and transformed after it has been loaded into the destination warehouse

10.] Comparison

Use Case

Summary

Recommended Format

Why?

Big Data Analytics/OLAP	Parquet	Fastest reads, best columnar compression
Data Ingestion/Streaming	Avro	Best writes, handles schema evolution
Data exchange/Small Data	CSV	Optimized indexes for Hadoop

Section B Artificial Intelligence & Machine Learning

11.] Artificial Intelligence (AI) is the science of creating computer systems capable of performing tasks that typically require human intelligence such as reasoning, learning, problem-solving and understanding language.

Narrow AI (Weak AI)

→ is designed and trained to perform a specific, limited task or narrow range of tasks: Example → Apple's Siri

General AI (Strong AI)

→ is a theoretical form of artificial intelligence that possesses the ability to understand, learn and apply knowledge to solve any intellectual problem, just as a human can: Example → A hypothetical, human-level AI

12.] Machine Learning (ML) is a subset of artificial intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Machine Learning vs Traditional Programming

- Traditional programming → A human programmer defines specific rigid rules (if then statements) to process input data and produce output
- Machine Learning → An algorithm is fed large amount of data, which it uses to infer the rules and logic themselves.

1. Supervised Learning → The model is trained on labeled data - input data that is already paired with the correct output.

2. Unsupervised Learning → The algorithm analyzes unlabeled data to find hidden patterns, structures, or relationships.

3. Reinforcement Learning → An agent learns to make decisions by interacting with an environment, receiving rewards for positive actions and penalties for negative ones.

13.] Deep Learning is a subset of machine learning (within AI) that uses multi-layered artificial neural networks to automatically learn complex patterns from vast, unstructured data.

• Artificial Intelligence (AI) → the overarching concept of simulating human intelligence in machines.

• Machine Learning (ML) → A subset of AI where machines learn from data rather than explicit programming.

• Deep Learning (DL) A specialized subset of ML that uses "deep" neural networks (many layers) to automatically find patterns without needing manual feature engineering.

→ Neural networks are computing systems inspired by the biological neural networks in the human brain.

14.] Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that enables computers to understand, interpret, and generate human language - both written and spoken.

1. Virtual Assistants & chatbots

2. Autocorrect and Predictive Text

3. Machine Translation

15.] Large Language Models (LLMs) are AI systems trained on vast datasets to understand and generate human-like text with examples including GPT-4o,

Gemini, LLaMa and Claude

Example → GPT-4o (OpenAI), Gemini 1.5 (Google)

16] Generative AI is a type of artificial intelligence that can create original content - including text, images, video, audio, and software code - in response to user prompts.

- ChatGPT (by OpenAI)

- DALL-E 3 (by OpenAI)

- GitHub Copilot (by GitHub)

17.] A training dataset is the initial information - such as text, images or data points - used to teach an AI model to recognize patterns, make predictions, or generate content (poor data leads to unreliable, biased and inaccurate AI behavior.)

- Accuracy & Performance → high-quality data ensures the model generalizes well to new, real-world scenarios rather than just memorizing input

18.] Computer vision is a field of artificial intelligence (AI) that enables machines to identify, analyze and interpret visual data from the world such as images and video mimicking human vision

- Data Input: Cameras and sensors act as the "eyes", capturing raw pixel data

Real world use → Autonomous Vehicles

Healthcare Imaging Diagnosis

19.]

Section C APIs, Tools & Developer Concepts

19.] An API is a set of rules and definitions that allows two software applications to talk to each other

Simple Real-world Analogy: The Restaurant

- You (The customer): You are the application or user requesting data

• The Kitchen (The Server) This is the system that prepares your food (data / functionality)

APIs Important Programming

• Accelerated Development (save time)

• Integration and Interoperability

20.] An API key is a unique, randomly generated alphanumeric string that serves as a digital credential to identify and authorize an application or user requesting access to an API

Purpose → Authentication & Identification

Access Control

Usage Monitoring

Rate Limiting

Storage and Protection → Never Hardcode

Use Environment Variables

Risk of Public Exposure → Unauthorized Access

Financial Loss

Service Disruption

21.] A REST API is an application programming interface that follows specific architectural design principles known as the REST style

Stands For → Representational State Transfer

HTTP Methods

• GET (Read) → Retrieves data from a specified resource without modifying it

• POST (Create) → Sends data to the server to create a new resource

• PUT (Update / Replace) → Updates an existing resource by replace its entire representation with new data

• PATCH (Partial Update) → Applies partial modifications to a resource instead of replacing the whole thing

• DELETE (Delete) - Removes a specific resource from the server

2.2] JSON is a lightweight, text-based data interchange format used to store and exchange structured data between systems.

JSON is Commonly Used for APIs

• Human Readable

• Lightweight

• Ease of Parsing

Simple JSON Object Example

→ A JSON object is a collection of key-value pairs enclosed in curly braces {}

23] Prompt engineering is the art and science of designing, refining and optimizing inputs (prompts) to guide generative AI models toward producing accurate, relevant, and high-quality outputs

24] A map() function is a higher-order function that applies a given function to each element of an iterable (such as a list, tuple or string)

Key Features of map() in Python

• Syntax

• Lazy Evaluation

Example Squaring a list of Numbers Using lambda

python List of numbers

numbers = [1, 2, 3, 4, 5]

25] Cloud computing is the on-demand delivery of IT resources - such as servers, storage, databases and software over the internet with pay-as-you-go-pricing

• Infrastructure as a Service (IaaS)

• Platform as a Service (PaaS)

• Software as a Service (SaaS)

26] A Version Control system (VCS) is software that

tracks and manages changes to code or files over time

Git has become the industry standard used by over 82% of developers [2, 11, 42]

27.] A Jupyter Notebook is an open-source, web-based interactive computing environment that allows users to create and share documents containing live code.

Most Common Programming language

→ Python

28.] Python is an interpreted, high-level general-purpose programming language developed by Guido van Rossum in 1991. It is widely considered one of the most popular languages, due to its simplicity, versatility and massive ecosystem

- Pandas
- Numpy
- Matplotlib
- Scikit learn

29.]

Section D Data Ethics Governance & Concerns

29.] The principle that individuals have control over their personal information, including how it is collected, stored, shared and used by organizations

- Prevents Misuse & Fraud
- Build Trust & Reputation

30.] Data ethics is a branch of ethics that evaluates the moral principles and guidelines governing the collection, processing and usage of data

- Data Privacy Violations and Unauthorized Scraping (Clearview AI)
- Algorithmic Discrimination and Racial Bias

- Manipulation of User behavior and Data Misuse
- 31] Data governance is a structured framework of policies, processes, roles and technology used to manage an organizations data, ensuring it is secure, high-quality and usable throughout its lifecycle
- Policies and Standards
- Roles and Responsibilities
- Security and Privacy
- 32] AI Bias occurs when AI systems produce systematically prejudiced or unfair results due to flawed training data, human assumptions
- Hiring and Recruitment (Gender and Age Bias)
- Criminal Justice and Law Enforcement (Racial Bias)
- 33] Data Engineer (The builder) Focuses on developing, maintaining and testing data infrastructure (pipelines, data warehouses)
- Data Scientist (The Modeler) focuses on advanced statistical analysis, machine learning and AI to forecast trends
- Data Analyst (The Interpreter) focuses on analyzing past data to answer business questions and creating visualizations
- 34] Business Intelligence (BI) refers to the procedural and technical infrastructure that collects, stores and analyzes data produced by a company's activities
- Microsoft Power BI
- Tableau

Section E Personal Reflection

- 35]. Section A Because it gives me the glue of what I will be learning
- 36]. Question 24 I used youtube to understand the concept
- 37] it was very bad

References

1. <https://mopivoh.com>
2. <https://nclbr.com>
3. <https://www.unesco.org>